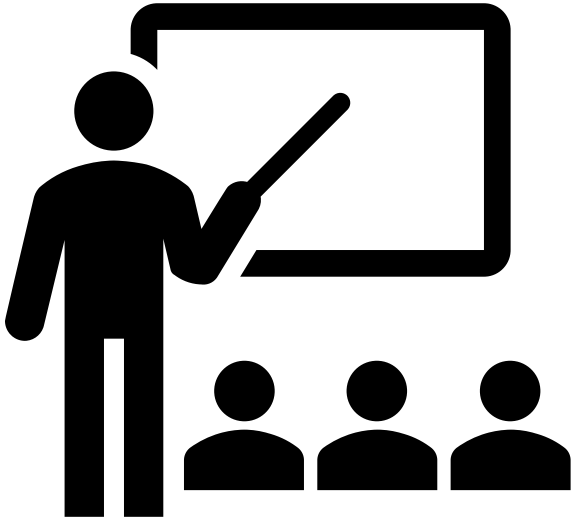


TUTORIAL DAY 2

Project Management

Sample Exam Paper

Duration	2 h 30 mins
Total questions	Answer 05 questions out of 06
Type	Closed book
Total Marks	100

Question Distribution	
Q1	• Project Management Overview and Concepts • Project Communications Management
Q2	• Project Scope Management • Project Integration Management
Q3	• Project Time Management
Q4	• Project Cost Management
Q5	• Project Human Resource Management • Project Quality Management
Q6	• Project Risk Management

Question 1

I. What does the term ‘project’ mean to you?

- A project is a temporary endeavor undertaken to create a unique product, service or result. The temporary nature of a project means that it has definite beginning or ending

Question 1

II. Get one example for each project and non-projects and explain how you differentiate from each other?

- Project Example: Building a prototype of a specific car model.
- Non-Project Example: Running an assembly line in a toy factory
- The primary difference lies in their nature and duration
  - ✓ Projects are temporary
  - ✓ Projects create a unique result
  - ✓ Projects are progressively elaborated. You learn more about the project as it moves along.
  - ✓ Non-projects are ongoing and repeatable

Question 1

III. Define the word “Project Owner” in relation to an IT development project

- Project Owner is defined as the individual or group who,
  - ✓ Provides the funding
  - ✓ Authorize the work.
  - ✓ Maintains ultimate responsibility
  - ✓ Defines Value

Question 1

IV. What type of standards and processes that you are willing to use in communication between foreign client and your company. ?

- Core Communication Processes.
  - ✓ Plan Communications Management
  - ✓ Manage Communications
  - ✓ Control Communications
- Communication Methods and Technologies
  - ✓ online collaboration tools
  - ✓ Formal Written Communication
  - ✓ Stakeholder Analysis
- Requirement Gathering Standards
  - ✓ Facilitated Workshops
  - ✓ Prototypes

Question 2

I. What does the term ‘Project Charter’ mean to you?

- A project charter is the foundational document that formally authorizes the existence of a project and officially authorizes the work to begin.
- It provides a high-level description of the project, including its purpose, business case, and summary milestone schedule
- Crucially, it grants the project manager the authority to use company resources (such as people, money, and equipment) and to assign work to team members for the duration of the project
- It acts as the formal announcement that a decision has been made to move forward with the project.

Question 2

III. What are the benefits of using Project Charter in Software Management ?

- Gives the project manager the necessary power to take control of resources and spend the project budget.
- Provides the formal "boss" status needed to put developers to work on project tasks.
- It serves as a tool to get all stakeholders on the same page regarding the project's high-level goals and requirements before detailed planning begins.
- Helps to understand why the company is investing in the product.

Question 2

III. List Down and Explain Triple Constraints in Project management.?

- **Scope**: All of the work that must be done to produce the project's deliverables.
- **Time** (Schedule): The requirement to finish the project work within a predefined timeline or by specific milestones.
- **Cost** (Budget): The limitation to stay within a specific budget and manage project investments correctly.
- Quality
- Resources.
- Risk.

Question 2

IV. What is a Work Breakdown Structure and how does it help in project scope management?

- A Work Breakdown Structure (WBS) is a graphical and hierarchical representation of all the work that must be performed to complete a project. It breaks down major deliverables or project phases into smaller, more manageable components called work packages.
- How it helps in Project Scope Management:
  - ✓ Defines Total Work
  - ✓ Prevents "Scope Creep“
  - ✓ Creates the Scope Baseline
  - ✓ Facilitates Estimation and Control

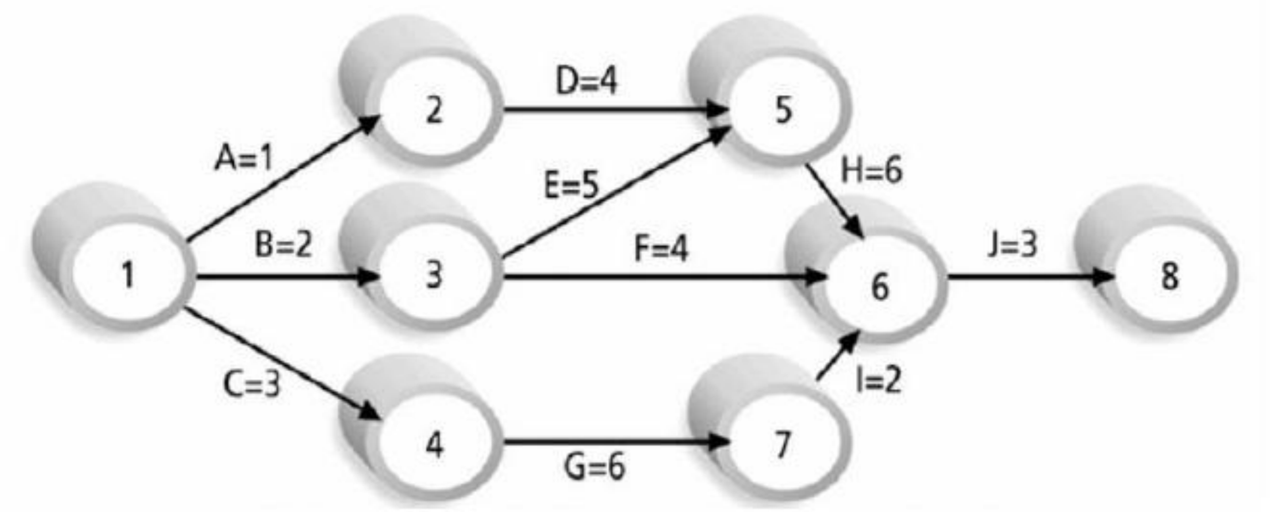
Question 3

I. What do you mean by Project Milestone?

- In project management, a milestone is a significant point or an important checkpoint used to track the progress of a project
- Unlike tasks or activities, milestones do not typically have a duration; they represent the successful completion of a major piece of work or a specific requirement.

Question 3

II. Using the following AOA network diagram, find the Critical Path. All duration is in days?

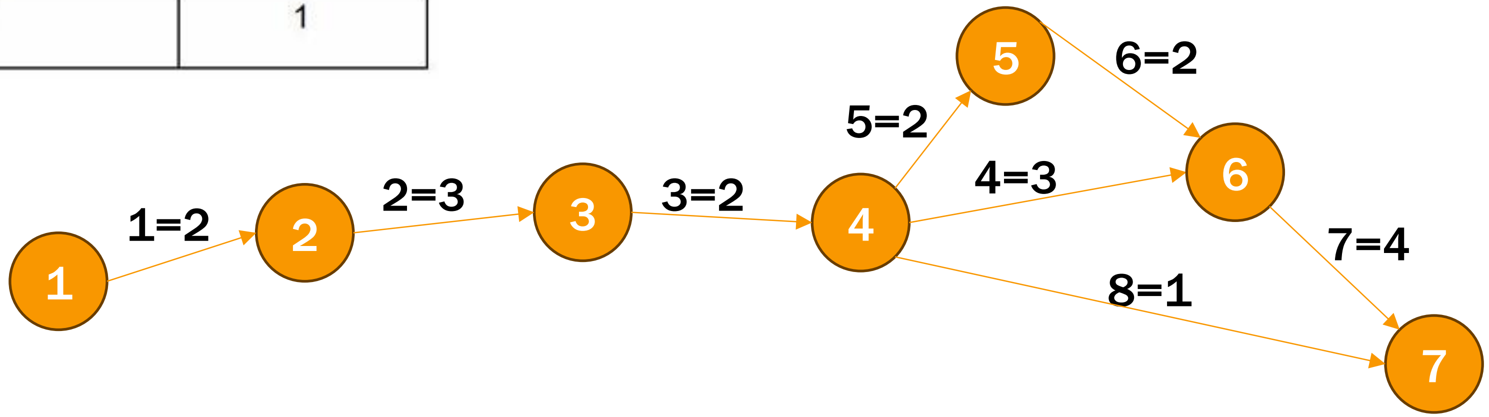


A > D > H > J	1+4+6+3	
B > E > H > J	2+5+6+3	
B > F > J	2+4+3	
C > G > I > > J	3+6+2+3	

Question 3

III. Draw the AOA using below mentioned data. ?

Project start date: 12 June 2015			
Task Identifier	Task Description	Predecessor Task(s)	Time (days)
1	Establish project	-	2
2	Establish customer requirements	1	3
3	Produce software specification documents	2	4
4	Write test plans	3	1
5	Write code	3	2
6	Developer testing	5	2
7	System testing	4, 6	4
8	Write customer documentation	3	1



Question 4

I. What is the main difference between Direct Cost and Indirect Cost.

- Direct Costs: These are expenses that can be directly attributed to a specific project activity or the production of project deliverables, such as team labor or materials.
- Indirect Costs: These are costs shared across the organization and not linked to a single project, such as office rent, utilities (electricity), or corporate administrative support.

Question 4

- **Budget at Completion (BAC):** \$106,000,000
- **Actual Cost (AC):** \$70,000,000
- **Actual % Complete:** 70% (or 0.70)
- **Time Elapsed:** 10 weeks
- **Total Planned Duration:** 14 weeks

i. Earned Value (EV)

- EV = BAC x Actual % complete.
  = 106000000 x 0.70

ii. Budget of Completion

- BAC = \$106000000

iii. Schedule Variance (SV)

- SV = EV - PV

Planned % complete = (10 weeks/14 weeks) x 100

PV = BAC x Planned % Complete

Question 4

- **Budget at Completion (BAC):** $106,000,000
- **Actual Cost (AC):** $70,000,000
- **Actual % Complete:** 70% (or 0.70)
- **Time Elapsed:** 10 weeks
- **Total Planned Duration:** 14 weeks

iv. Schedule performance index (SPI)

- SPI = EV / PV.

ii. Cost Variance (CV)

- CV = EV - AC

iii. Cost performance index (CPI)

- CPI = EV / AC

Question 4

III. Briefly explain 02 out of the following terms of Cost Management.

- **Tangible costs**: These are expenses that can be easily measured in specific monetary units, such as the cost of labor, materials, or hardware. They represent physical assets that contribute to business value, like desks and chairs.
- **Intangible costs:** These refer to costs that are difficult to quantify in dollar terms, such as the impact on employee morale, brand reputation, or the loss of intellectual property. Business value is comprised of both tangible items and these intangible elements.
- **Sunk cost:** This refers to money that has already been spent in the past on a project. According to standard project management principles, sunk costs should be ignored when making decisions about whether to continue, modify, or terminate a project, as that money cannot be recovered.

Question 5

I. What does the term ‘Project Quality’ mean to you?

- In project management, quality means ensuring that you build exactly what you said you would build and doing so as efficiently as possible
- It is defined by two primary concepts:
  - ✓ Conformance to Requirements
  - ✓ Fitness for Use

Question 5

II. Write down 03 different techniques that we are using in Software Testing

- Unit Testing
- Code Reviews.
- Design of Experiments

Question 5

III. What does the term ‘Project Human Resource Management’ mean to you?

- Project Human Resource Management includes the processes required to make the most effective use of the people involved with a project
- It involves more than just putting a team together; it requires careful planning for staffing needs, negotiating for the best resources, establishing a positive work environment, keeping the team motivated, and resolving any conflicts that arise during the project.

Question 5

IV. Write down 05 process of Tuckman Model in developing the project team?

- Forming.
- Storming.
- Norming
- Performing.
- Adjourning

Question 5

V. Draw down Maslow's Hierarchy of Needs

Question 6

I. What does the term ‘Project Risk Management’ mean to you?

- Project Risk Management is defined as the art and science of identifying, analyzing, and responding to risk throughout the life of a project
- It is a systematic process designed to minimize the impact of potential negative events (threats) while maximizing the impact of potential positive events (opportunities) to ensure project objectives are met
- Rather than just being reactive, it serves as a form of preventative insurance for the project.

Question 6

II. Write down 03 differences between Qualitative risk analysis and Quantitative risk analysis.

- Qualitative analysis prioritizes risks based on subjective judgment using predefined scales (e.g., low, medium, high). In contrast, quantitative analysis uses numerical techniques to assign exact values to the probability and impact of risks.
- Qualitative analysis is used to rank and prioritize risks to determine which ones need immediate attention. Quantitative analysis focuses on providing the hard numbers (such as cost in dollars or time in days) to back up those initial judgments.
- Qualitative analysis is typically performed on almost all projects to categorize risks. Quantitative analysis is more complex and generally reserved for large, complex projects involving advanced modeling like Monte Carlo simulations or Decision Tree analysis.

Question 6

III. Write down 06 Project Risk Management process.

- Risk Management Planning.
- Risk Identification.
- Qualitative Risk Analysis.
- Quantitative Risk Analysis.
- Risk Response Planning.
- Risk Monitoring and Control

Question 6

IV. Write down 03 different risk identification techniques/tools.

- Brainstorming.
- Delphi Technique.
- SWOT Analysis.